Lecture 1

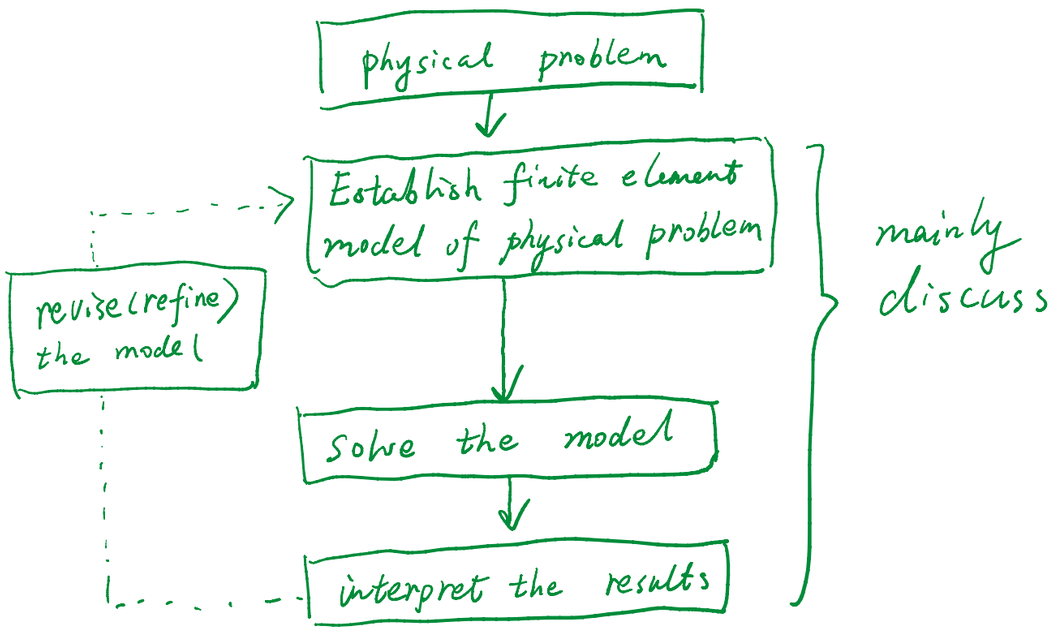
Sunday, January 27, 2019 2:20 PM

Techniques:

SAP: structural Analysis Program

ADINA: Automatic Dynamic Incremental Nonlinear Analysis.

Finite Element Solution Process



Basic Concepts

- idealization of system
- formulation of equilibrium equations
- solution of equations
- interpretation of results.

Systems

DISCRETE	CONTINUOUS
- response is described by variables at a FINITE number of points.	- response is described by variables at an INFINITE number of points
- set of algebraic equations	- set of differential equations

Problem Types

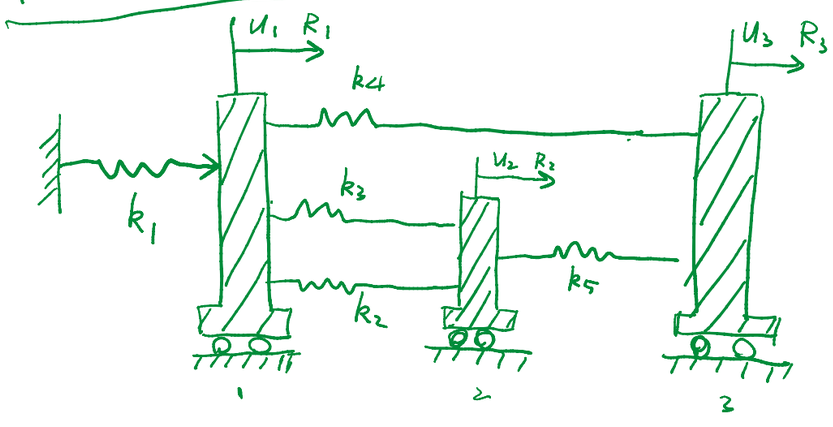
- Steady state (statics)
- Propagation (dynamics)
- Eigenvalue

Analysis of Discrete Systems.

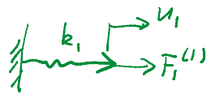
Steps:

- system idealization into elements → Don't need this step in continuous systems.
- evaluation of element equilibrium requirements
- element assemblage. 集合
- solution of response

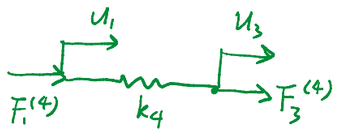
Problem 1 steady state



1. Element



$k_1 u_1 = F_1^{(1)}$



$$k_4 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_3 \end{bmatrix} = \begin{bmatrix} F_1^{(4)} \\ F_3^{(4)} \end{bmatrix}$$

. . .

2. Element interconnection requirements

$$F_1^{(1)} + F_1^{(2)} + F_1^{(3)} + F_1^{(4)} = R_1$$
$$F_2^{(2)} - F_2^{(3)} + F_2^{(5)} = R_2$$
$$F_3^{(4)} + F_3^{(5)} = R_3$$

$$\Rightarrow KU = R \quad \begin{cases} U^T = [u_1 \ u_2 \ u_3]^T \\ R^T = [R_1 \ R_2 \ R_3]^T \\ K = \begin{bmatrix} \sum_1^4 k_i & -k_2-k_3 & -k_4 \\ -k_2-k_3 & k_2+k_3+k_5 & -k_5 \\ -k_4 & -k_5 & k_4+k_5 \end{bmatrix} \end{cases}$$

$\rightarrow$ $\boxed{\text{governing}}$ the system

$(K = \sum_i k_i)$ e.g. $K_1 = \begin{bmatrix} k_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $K_2 = \begin{bmatrix} k_2 & -k_2 & 0 \\ -k_2 & k_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\Rightarrow$ direct stiffness method.

3. Another solution: variational approach.

$$\Pi = U - W$$

U: strain energy of system

W: total potential of the loads

- Equilibrium equations:

$$\frac{\partial \Pi}{\partial u_i} = 0 \quad \text{for } \forall u_i$$

for the above problem:

$$U = \frac{1}{2} U^T K U$$
$$W = U^T R$$
$$\Rightarrow \Pi = U - W = \frac{1}{2} U^T K U - U^T R$$

$$\frac{\partial \Pi}{\partial U} = 0 \Rightarrow KU - R = 0 \Rightarrow KU = R$$

Problem 2: Propagation problems

main characteristic: response changes with time

$\Rightarrow$ add d'Alembert forces

$$KU(t) = R(t) - M\ddot{U}(t)$$

Example: system of rigid carts

$$M\ddot{U} + KU = 0$$

Let: $U = \phi \sin \omega(t-\tau)$

$\Rightarrow -\omega^2 M\phi \cancel{\sin \omega(t-\tau)} + K\phi \cancel{\sin \omega(t-\tau)} = 0$

$\Rightarrow K\phi = \omega^2 M \Rightarrow$ EVP

Problem 3: Eigenvalue problems (EVP)

$$Av = \lambda Bv \qquad \text{e.g. } K\phi = \omega^2 M\phi$$

$A, B \in S_{n\times n}$ $\quad$ n eigen pairs